Vol 14 Chapter 3 — Shannon Channel Capacity

Keeper (author pass — deep math/physics revision)

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Chapter 3 — Shannon Channel Capacity

Level 1 — one sentence

Shannon 1948 founded information theory with the channel capacity theorem: any channel has a maximum reliable transmission rate $C = \max_{p(x)} I(X;Y)$ (bits/symbol), achievable in principle via optimal coding — and BST predicts a substrate channel capacity set by BST primaries and the Koons-tick clock.

Level 2 — graduate-physicist precision

3.1 Entropy and mutual information

Shannon entropy of source X with distribution p:

$$H(X) = -\sum_x p(x) \log_2 p(x) \text{ bits}$$

Conditional entropy:

$$H(Y|X) = \sum_x p(x) H(Y|X = x)$$

Mutual information:

$$I(X;Y) = H(Y) - H(Y|X) = H(X) - H(X|Y) = H(X) + H(Y) - H(X,Y)$$

Measures how much knowing X tells you about Y.

3.2 Channel capacity

For channel with transition probabilities $p(y|x)$:

$$C = \max_{p(x)} I(X;Y)$$

Shannon's noisy channel coding theorem: for any rate $R < C$, there exists a code with vanishing error probability as block length $n \to \infty$.

Converse: for $R > C$, error probability is bounded away from 0.

3.3 Standard channel examples

- Binary symmetric channel (BSC) with error probability p: $C = 1 - H_2(p)$ where $H_2(p) = -p\log_2 p - (1 - p)\log_2(1-p)$
- Additive white Gaussian noise (AWGN): $C = (1/2)\log_2(1 + S/N)$ bits per use (Shannon-Hartley)
- Erasure channel with erasure probability ϵ: $C = 1 - \epsilon$

3.4 BST substrate channel

Substrate channel capacity per Koons tick:

$$C_{\text{substrate}} = g = 7 \text{ bits/tick}$$

based on alphabet $GF(2^g) = GF(128)$ (Ch 2).

Per-second capacity:

$$C_{\text{substrate}}/t_K = 7/10^{-120} = 7 \cdot 10^{120} \text{ bits/s}$$

— astronomical. The substrate is computing at extraordinary rates; what we observe as physics is the coarse-grained output.

3.5 Substrate Channel Capacity Bound (SCCB — Casey-named CANDIDATE, 2026-05-24)

Candidate principle (Casey decision 2026-05-24: candidate-only, pending mechanism work): no observable can be encoded at higher rate than substrate channel capacity $C_{\text{substrate}} = g = 7$ bits/Koons-tick $\approx 7 \cdot 10^{120}$ bits/s.

Implications if ratified: - Cosmological information bounds (Bekenstein, holographic) reduce to SCCB at specific boundary conditions - Black hole entropy is substrate-channel-capacity boundary effect

Status: I-tier hypothesis. Mechanism path: Lyra to formalize via substrate-Hilbert-space dimension counting (overlaps SP-31). Falsifier: identify physical encoding rate above $C_{\text{substrate}}$ in any observable setting.

3.6 K-audit anchors

- Paper #122: Information Substrate
- Vol 4 Ch 9 (holographic bounds)

Level 3 — 5th-grader accessibility

Shannon 1948 founded information theory. Entropy = how surprising the message is (bits). Channel capacity = max reliable rate; noisy channel theorem says you can get arbitrarily close to it with the right coding. BSC with error p: $C = 1 - H_2(p)$. Gaussian channel: $C = (1/2)\log(1 + S/N)$. BST substrate channel: $g = 7$ bits/Koons-tick $= 7 \cdot 10^{120}$ bits/s. Substrate is computing at extraordinary rates; physics is the coarse-grained output.

What comes next

Chapter 4 develops Nyquist sampling + Koons tick.

Where to look this up

- Shannon 1948
- Cover-Thomas, *Elements of Information Theory*
- BST: Paper #122; Vol 4 Ch 9